

Programming Practice Questions

1. State the order of evaluation of the operators in each of the following C statements and show the value of x after each statement is performed.

a) $x = 7 + 3 * 6 / 2 - 1;$

b) $x = 2 \% 2 + 2 * 2 - 2 / 2;$

c) $x = (3 * 9 * (3 + (9 * 3 / (3))));$

2. Write a program that asks the user to enter two numbers, obtains them from the user and prints their sum, product, difference, quotient and remainder.

3. Write a program that prints the numbers 1 to 4 on the same line. Write the program using the following methods. a) Using one printf statement with no conversion specifiers. b) Using one printf statement with four conversion specifiers. c) Using four printf statements.

4. Write a program that asks the user to enter two integers, obtains the numbers from the user, then prints the larger number followed by the words "is larger." If the numbers are equal, print the message "These numbers are equal." Use only the single-selection form of the if statement.

5. Write a program that inputs three different integers from the keyboard, then prints the sum, the average, the product, the smallest and the largest of these numbers. Use only the single-selection form of the if statement.

6. Write a program that reads in the radius of a circle and prints the circle's diameter, circumference and area. Use the constant value 3.14159 for π . Perform each of these calculations inside the printf statement(s) and use the conversion specifier %f.

7. Write a program that reads an integer and determines and prints whether it is odd or even.

8. Write a program that reads in two integers and determines and prints if the first is a multiple of the second.

9. Create a BMI calculator application that reads the user's weight in pounds and height in inches (or, if you prefer, the user's weight in kilograms and height in meters), then calculates and displays the user's body mass index. Also, the application should display the following information from the Department of Health and Human Services/National Institutes of Health so the user can evaluate his/her BMI:

BMI Values

Underweight: less than 18.5

Normal: between 18.5 & 24.9

Overweight: between 25 & 29.9

Obese: 30 or greater

Formula for calculating BMI is as follow:

$$\text{BMI} = \frac{\text{weight in kilograms}}{\text{height in meters} * \text{height in meters}}$$

Or

$$\text{BMI} = \frac{\text{weight in pounds} * 703}{\text{height in inches} * \text{height in inches}}$$

10. Write a program that reads an integer and determines and prints how many digits in the integer are 5s.
11. Write a program that sums a sequence of integers. Assume that the first integer read with scanf specifies the number of values remaining to be entered. Your program should read only one value each time scanf is executed. A typical input sequence might be **5 100 200 300 400 500** where the 5 indicates that the subsequent five values are to be summed.
12. Write a program that calculates and prints the average of several integers. Assume the last value read with scanf is the sentinel 9999. A typical input sequence might be **10 8 11 7 9 9999** indicating that the average of all the values preceding 9999 is to be calculated.
13. Write a program that finds the smallest of several integers. Assume that the first value read specifies the number of values remaining.
14. Write a program that calculates and prints the sum of the even integers from 2 to 30.
15. Write a program that calculates and prints the product of the odd integers from 1 to 15.
16. The factorial function is used frequently in probability problems. The factorial of a positive integer n (written n! and pronounced "n factorial") is equal to the product of the positive integers from 1 to n. Write a program that evaluates the factorials of the integers from 1 to 5. Print the results in tabular format.
17. Write a program that asks user to give two numbers and an operator (+, -, *, /) to the program and perform the appropriate operation based on the provided operator.
18. Write a program that prints the following patterns separately, one below the other. Use for loops to generate the patterns. All asterisks (*) should be printed by a single printf statement of the form printf("%*");

A)

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* *
* * *
* * * *
* * * * *
```

B)

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* * * * *
* * * *
* * *
* *
*
```

C)

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      *
      * *
      * * *
      * * * *
      * * * * *
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D)

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* * * * *
* * * *
* * *
* *
*
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19. Write a program that asks user to give number of students in the class. Maximum number of students in a class can be 10. Then calculate average GPA of the class by asking GPA of each student from the user and displays the Class Average.
20. Write a program that checks whether a given number is prime number or not.

21. Write a program that take two integers as input and returns LCM of the given two numbers.
22. Write a program that calculates power of a number where number & power both will be provided by the user.
23. Write a program that calculates the sum of all the digits of a number provided by the user.
24. Write a program to find the factorial of a given number.
25. Write a program that print the multiples of a number from 'n' to 'm' where 'n' is the starting point and 'm' is the ending point.

For example

Enter a number: 2

Enter the starting point: 6

Enter the ending point: 18

Multiples: 6, 8, 10, 12, 14, 16, 18